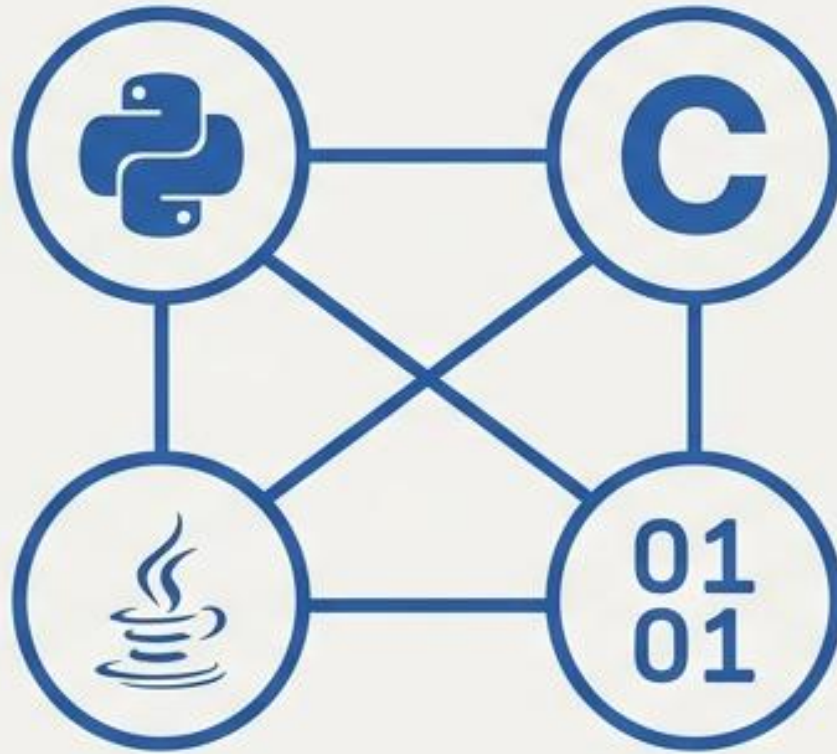


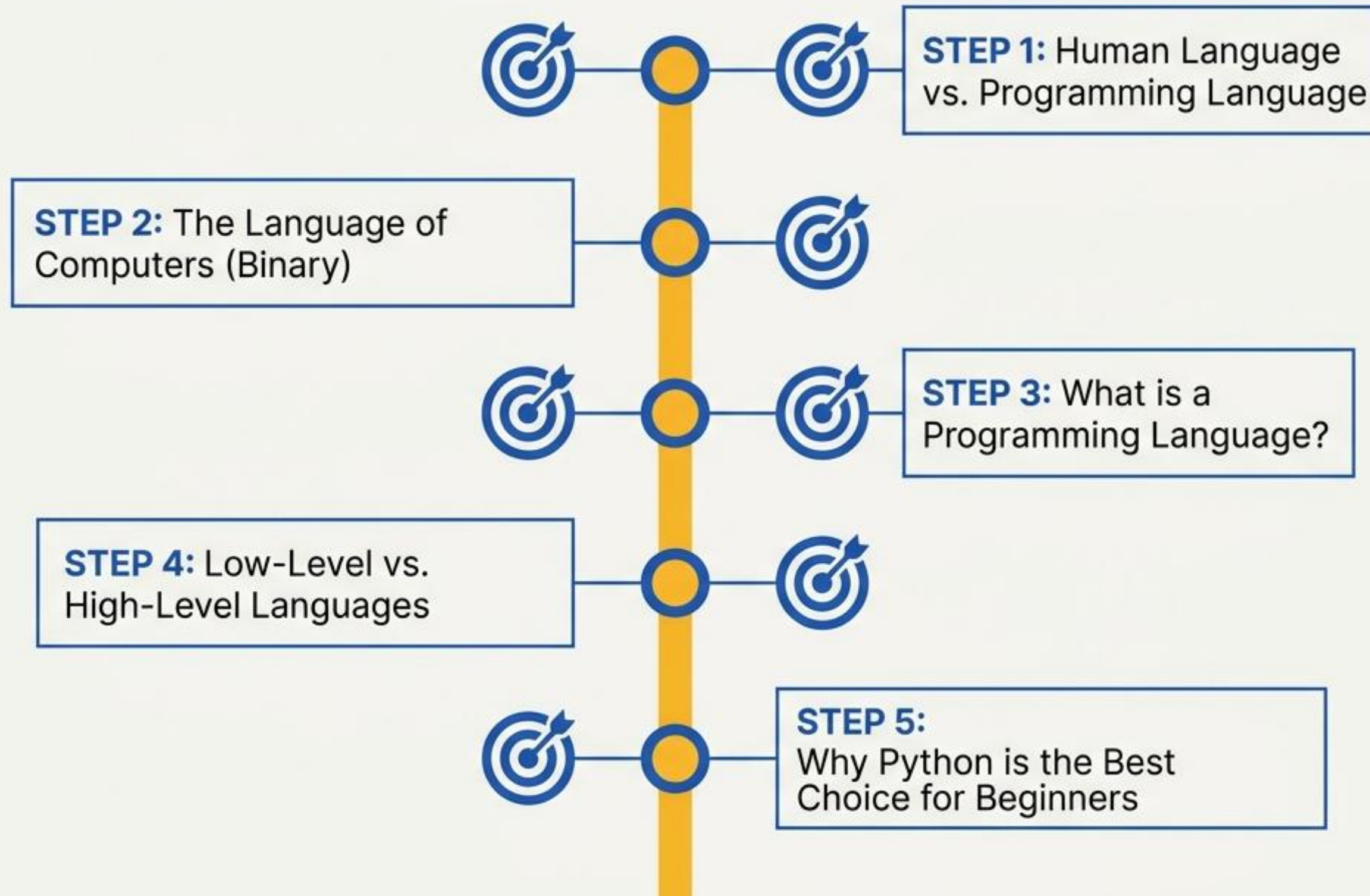
Programming Languages

How Humans Communicate with Computers



Welcome to **Lesson 2**. Today we will learn exactly how computers **understand our instructions**.

Mapping Your Learning Journey



How Do Humans Communicate?

Humans communicate using languages to share ideas and complex information.

Examples of human languages:

English

Hindi

Urdu

Arabic



Can Computers Understand Human Language?

DEFINITION: Binary Language

The language of computers consisting entirely of exactly two symbols: 0 and 1.

Computers cannot understand English or human emotions directly.

They only understand Binary Language.



What is a Programming Language?

They act as translators, converting our instructions into a format the computer understands.



DEFINITION: Programming Language

A formal language used to write precise instructions that a computer can execute.

Comparing Human and Machine Communication

Human Language	Programming Language
 Used by Humans	 Used by Computers
 Examples: English, Hindi, Urdu	 Examples: Python, Java, C++

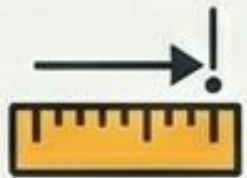
Why Can't We Write Programs in English?



Computers do not understand emotions.



Computers never guess.



Every instruction must be exact.



Even a small mistake can produce errors.

Real-Life Example Callout



Show me the nearest hospital.



Programming works behind the scenes.



Calculated precise route instructions.

Quick Check #1

Which language does the computer understand directly?

[A] English

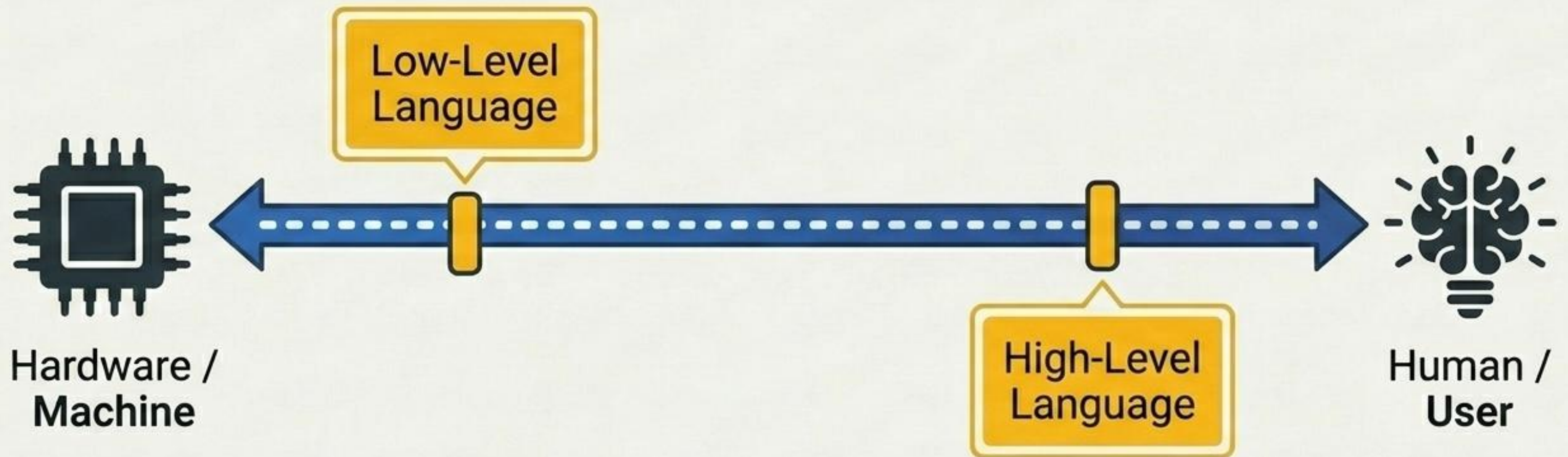
[B] Hindi

[C] Binary

[D] Python

The Two Categories of Programming Languages

Programming languages are split into two major categories based on how close they are to raw machine language versus human language.



Low-Level Languages Speak to the Machine

DEFINITION: Low-Level Language

A programming language that is very close to raw machine language.

- ✓ Faster execution speed.
- ✓ Difficult for humans to read and write.
- ✓ Machine dependent (tied to specific hardware).
- ✓ Used primarily in core system programming.



Example: Assembly Language

High-Level Languages Speak to the Human

DEFINITION: High-Level Language

A programming language designed to be easy for humans to read, write, and understand.

- ✓ Easy, simple syntax to learn.
- ✓ Portable across different machines.
- ✓ Used in modern software development.

Examples:



Python

Java

C#

JavaScript

Matrix: Low-Level vs. High-Level

	Low-Level Language	High-Level Language
Difficulty	Difficult to learn	Easy to learn
Speed	Extremely Fast	Slightly Slower
Portability	Machine Dependent	Portable anywhere
Syntax	Complex & Technical	Simple & English-like

Quick Check #2

Which of these is easier for beginners to learn?

[A] Machine Code

[B] Assembly Language

[C] Python

[D] Binary

Why Python is the Ultimate Starting Point

DEFINITION: Python

A high-level, interpreted programming language designed to be simple, readable, and incredibly easy to learn.



Simple,
English-like
syntax



Requires
less code to
do more



Beginner
friendly



Supported by
a huge
community



Free and
Open Source



Where is Python Used?



**Artificial
Intelligence**



**Machine
Learning**



**Data
Science**



**Web
Development**



Automation



**Cyber
Security**



Robotics



**Game
Development**

Why Most Universities Choose Python First

- ✓ **Easy to learn**
- ✓ **Focuses on programming logic**
- ✓ **Less confusing syntax**
- ✓ **Excellent career opportunities**

Trusted by Industry Leaders



Which language is widely recommended for beginners?

A Assembly

B Python

C Binary

D Machine Code

Did You Know?



Python was created by Guido van Rossum.

First released in 1991.

Today, it is one of the most popular programming languages in the world.

How We Communicate with Computers

Programming Language: A tool that helps humans communicate instructions to computers.

Low-Level Language

Examples: Binary, Machine Code.

Traits: Computers understand this directly. Difficult for humans to read or write.

High-Level Language

Examples: Python.

Traits: Closer to human language. Beginner-friendly and easy for humans to write.

Key Takeaways

- ✓ Computers understand only Binary Language.
- ✓ Python is a High-Level Language.
- ✓ Python is beginner-friendly and widely used across industries.
- ✓ Learning Python is the first step toward becoming a programmer.

